Final exam · Unit 3 · Reduction of order · 2.5 points · 50 min · Standard exam level

P15 — x y'' - (x+2) y' + 2y = 0 with the known solution e^x: a polynomial second solution by reduction of order, the forced equation with x^3 e^x, and initial-value problems at the singular point x = 0

Consider, for $x > 0$, the equation

$$x\,y'' - (x+2)\,y' + 2y = 0.$$

(a) Check that $y_1(x) = e^x$ is a solution. Using $y = v(x)\,e^x$, find a second solution y_2, linearly independent of y_1, and compute the Wronskian $W(y_1, y_2)$.

(b) Find the general solution of

$$x\,y'' - (x+2)\,y' + 2y = x^3 e^x, \qquad x > 0,$$

again with the substitution $y = v\,e^x$.

(c) The solutions of (b) extend smoothly to $x = 0$. Show that all of them satisfy $y'(0) = y(0)$. What does this say about the initial-value problems $y(0) = 1,\ y'(0) = 0$ and $y(0) = 1,\ y'(0) = 1$? Why does this not contradict Picard's theorem?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $y(x)$, independent variable x, domain $x > 0$ (the leading coefficient x vanishes at $x = 0$).
- Operator: linear, second order, variable coefficients; (a) homogeneous, (b) non-homogeneous with $g = x^3 e^x$.
- One solution is known, so the order can be reduced: v' satisfies a first-order linear equation.

Phase 2. Canonical formulation

Dividing by x gives the standard form $y'' + p(x)y' + q(x)y = f(x)$ with

$$p(x) = -\frac{x+2}{x} = -1 - \frac{2}{x}, \qquad q(x) = \frac{2}{x}, \qquad f(x) = x^2 e^x.$$

p, q, f are continuous on $x > 0$, so every initial-value problem posed at $x_0 > 0$ has a unique solution there; at $x_0 = 0$ the hypotheses fail.

Part (a) · 1 point

Second solution by reduction of order and the Wronskian

Step 1. Check of y1

$y_1 = y_1' = y_1'' = e^x$: $xe^x - (x+2)e^x + 2e^x = (x - x - 2 + 2)e^x = 0$.

Step 2. Substitution

$y = ve^x$, product rule: $y' = e^x(v' + v)$, $y'' = e^x(v'' + 2v' + v)$. Substituting:

$$xe^x(v'' + 2v' + v) - (x+2)e^x(v' + v) + 2ve^x = e^x\big[xv'' + (2x - x - 2)v' + (x - x - 2 + 2)v\big] = e^x\big[xv'' + (x-2)v'\big]$$

So the homogeneous equation becomes $xv'' + (x-2)v' = 0$.

Step 3. First-order equation for w = v'

$xw' = -(x-2)w$, separable: $\dfrac{dw}{w} = \Big(-1 + \dfrac{2}{x}\Big)dx$. Integrating, $\ln|w| = -x + 2\ln x + c$, so $w = Kx^2e^{-x}$; take $K = 1$.

Step 4. Integrate w

By parts twice ($u = x^2$, $dv = e^{-x}dx$, $du = 2x\,dx$, $v = -e^{-x}$; then $u = x$, $du = dx$):

$$\int x^2e^{-x}dx = -x^2e^{-x} + 2\int xe^{-x}dx = -x^2e^{-x} + 2\Big(-xe^{-x} + \int e^{-x}dx\Big) = -(x^2 + 2x + 2)e^{-x}$$

Then $y = ve^x = -(x^2 + 2x + 2)$. Dropping the constant factor -1: $y_2 = x^2 + 2x + 2$. Check: $y_2' = 2x + 2$, $y_2'' = 2$: $2x - (x+2)(2x+2) + 2(x^2 + 2x + 2) = 2x - (2x^2 + 6x + 4) + 2x^2 + 4x + 4 = 0$.

Step 5. Wronskian

$$W = \begin{vmatrix} e^x & x^2 + 2x + 2 \\ e^x & 2x + 2 \end{vmatrix} = e^x(2x+2) - e^x(x^2 + 2x + 2) = -x^2e^x$$

It never vanishes on $x > 0$, so $\{y_1, y_2\}$ is a fundamental set. Abel's formula agrees: $W = C\exp\Big(-\int p\,dx\Big) = C\exp\Big(x + 2\ln x\Big) = Cx^2e^x$, with $C = -1$.

KEY RESULTS

- $xv'' + (x-2)v' = 0$, so $w = v' = x^2 e^{-x}$.
- $v = -(x^2+2x+2)e^{-x}$, hence $y_2 = x^2 + 2x + 2$ (a polynomial).
- $W(e^x,\ x^2+2x+2) = -x^2 e^x \neq 0$ for $x > 0$.

Part (b) · 1 point

General solution of the forced equation

Step 1. Reduced equation

From Step 2 of (a), $L[ve^x] = e^x\left[xv'' + (x-2)v'\right] = x^3 e^x$, so $xv'' + (x-2)v' = x^3$. With $w = v'$ and dividing by x:

$$w' + \left(1 - \frac{2}{x}\right)w = x^2$$

Step 2. Integrating factor

$\mu = \exp\int\left(1 - \frac{2}{x}\right)dx = \exp(x - 2\ln x) = \frac{e^x}{x^2}$. Check: $\mu' = \frac{e^x}{x^2} - \frac{2e^x}{x^3} = \mu\left(1 - \frac{2}{x}\right)$. Then $(\mu w)' = \mu w' + \mu' w = \mu\left[w' + (1 - 2/x)w\right] = \frac{e^x}{x^2}x^2 = e^x$.

Step 3. Integrate twice

$\mu w = e^x + C$, so $w = x^2 + Cx^2 e^{-x}$. Integrating, with the antiderivative of (a) Step 4:

$$v = \frac{x^3}{3} - C(x^2+2x+2)e^{-x} + D$$

and $y = ve^x$:

$$y = De^x - C(x^2+2x+2) + \frac{x^3 e^x}{3} = C_1 e^x + C_2(x^2+2x+2) + \frac{x^3 e^x}{3}$$

Step 4. Check of the particular solution

$y_p = \frac{1}{3}x^3e^x$: $y_p' = \left(x^2 + \frac{1}{3}x^3\right)e^x$, $y_p'' = \left(2x + 2x^2 + \frac{1}{3}x^3\right)e^x$. Then

$$xy_p'' - (x+2)y_p' + 2y_p = e^x\left[2x^2 + 2x^3 + \tfrac{1}{3}x^4 - \left(x^3 + \tfrac{1}{3}x^4 + 2x^2 + \tfrac{2}{3}x^3\right) + \tfrac{2}{3}x^3\right] = x^3e^x$$

Variation of constants with $f = x^2e^x$ and $W = -x^2e^x$ gives $y_p - 2e^x$, the same up to a homogeneous solution.

KEY RESULTS

- $w' + \left(1 - \dfrac{2}{x}\right)w = x^2, \mu = \dfrac{e^x}{x^2}, (\mu w)' = e^x.$
- $w = x^2 + Cx^2e^{-x}, v = \dfrac{x^3}{3} - C(x^2 + 2x + 2)e^{-x} + D.$
- $y = C_1e^x + C_2(x^2 + 2x + 2) + \dfrac{x^3e^x}{3}.$

Part (c) · 0.5 points

Initial conditions at the singular point x = 0

Step 1. Values at x = 0

$y = C_1e^x + C_2(x^2 + 2x + 2) + \frac{1}{3}x^3e^x$ gives $y(0) = C_1 + 2C_2$. $y' = C_1e^x + C_2(2x + 2) + \left(x^2 + \frac{1}{3}x^3\right)e^x$ gives $y'(0) = C_1 + 2C_2$. So $y'(0) = y(0)$ always. Directly: at $x = 0$ the equation reads $0 \cdot y''(0) - 2y'(0) + 2y(0) = 0$.

Step 2. The two initial-value problems

$y(0) = 1, y'(0) = 0$ violates $y'(0) = y(0)$: **no solution**. $y(0) = y'(0) = 1$ only gives $C_1 + 2C_2 = 1$: a one-parameter family, **no uniqueness**.

Step 3. Picard

Picard's theorem needs the standard form $y'' = f(x) - p(x)y' - q(x)y$ with coefficients continuous near x_0. At $x_0 = 0$, p and q blow up, so the theorem does not apply. Consistently, $W = -x^2e^x$ vanishes at $x = 0$: the matrix $\begin{pmatrix} y_1(0) & y_2(0) \\ y_1'(0) & y_2'(0) \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$ is singular, so the constants cannot be fitted to arbitrary data.

KEY RESULTS

- $y(0) = C_1 + 2C_2 = y'(0)$ for every solution.
- $y(0) = 1,\ y'(0) = 0$: no solution. $y(0) = y'(0) = 1$: infinitely many ($C_1 = 1 - 2C_2$, C_2 free).
- No contradiction: $p = -1 - 2/x$ and $q = 2/x$ are not continuous at $x = 0$ (equivalently $W(0) = 0$).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** as $x \to \infty$ the forced response $\frac{1}{3}x^3e^x$ dominates both e^x and the polynomial. As $x \to 0^+$ every solution is smooth, but only data on the line $y'(0) = y(0)$ are reachable.
- **Checks:** y_1, y_2, y_p substituted back (Steps (a)4 and (b)4); Abel's formula reproduces the Wronskian; $W \neq 0$ on $x > 0$.
- **Common mistakes in the exam:** (1) forgetting to divide by x before reading $p(x)$ for Abel's formula or variation of constants (with $f = x^2e^x$, not x^3e^x); (2) sign slips in $\int x^2e^{-x}dx$; (3) presenting $y_2 = -(x^2 + 2x + 2)e^{-x}$, i.e. v instead of $y = ve^x$; (4) claiming uniqueness at $x_0 = 0$ by quoting Picard without checking continuity of p and q.